

Computer Vision Architecture

Fundamentals, Hierarchies, and Real-World Perception

Technical Evaluation Presentation | Engineering & AI Research



Theoretical Foundations

Mathematical Image Formulations & Inverse Rendering

Core Mathematical Concepts

Multidimensional Image Tensors

Digital images are represented as multidimensional arrays where spatial grid coordinates map directly to intensity values across color channels:

$$\in \mathbb{H} \times \mathbb{W} \times \mathbb{C}$$

For RGB images, $C = 3$; for grayscale, $C = 1$. Downstream algorithms operate on continuous tensor gradients.

The Perspective Projection Model

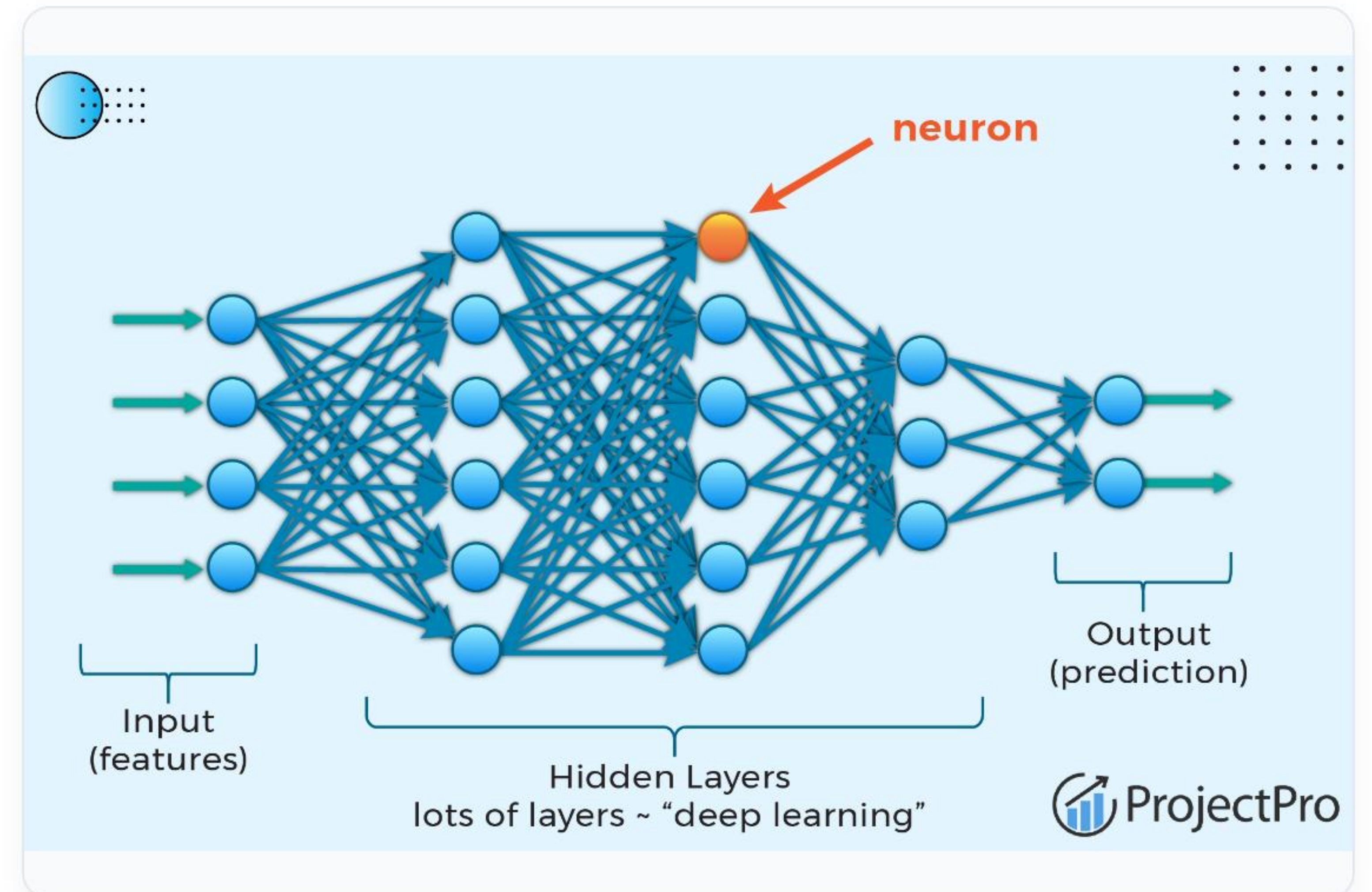
Computer Vision solves an ill-posed *inverse rendering problem*, reconstructing 3D scene properties from 2D pixel projections:

$$P_{2D} = K \cdot [R \mid t] \cdot P_{3D}$$

Where K encapsulates intrinsic camera parameters, and $[R \mid t]$ denotes extrinsic rotation and translation transformations.

Key Features & Capabilities

-  **Object Recognition & Bounding Boxes:** Simultaneous classification and localization identifying "what" and "where" within visual fields.
-  **Semantic & Instance Segmentation:** Classifying pixels individually to demarcate precise visual boundaries (e.g., U-Net, Mask R-CNN).
-  **Optical Flow & Spatial Tracking:** Estimating frame-to-frame pixel velocity vectors to track object trajectories through time.
-  **3D Scene Reconstruction:** Estimating depth maps via stereo disparity and Structure-from-Motion (SfM) algorithms.



Levels of Computer Vision



1. Low-Level Vision

Image-to-image transformations focused on raw pixel processing. Includes Gaussian filtering, edge detection (Canny/Sobel), and histogram equalization to suppress noise.



2. Mid-Level Vision

Extracting structural primitives and geometry. Involves region segmentation, contour detection, and keypoint matching (SIFT/ORB) to isolate visual primitives.



3. High-Level Vision

Cognitive scene understanding using Deep Neural Networks (CNNs/Transformers). Delivers multi-class detection, behavioral intent estimation, and context-aware reasoning.



Real-World Case Study

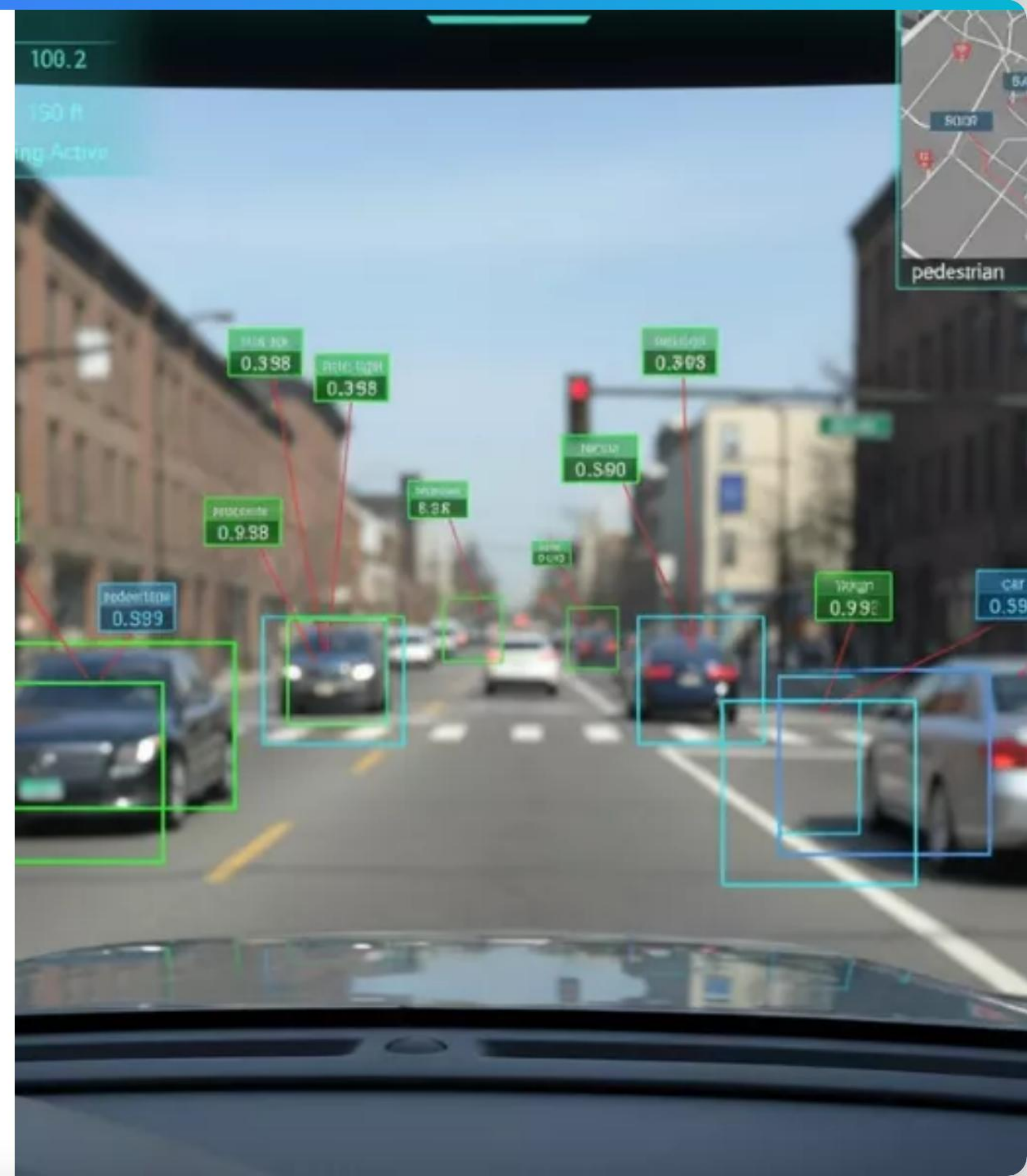
Autonomous Vehicle Perception System Architecture

Autonomous Perception Problem

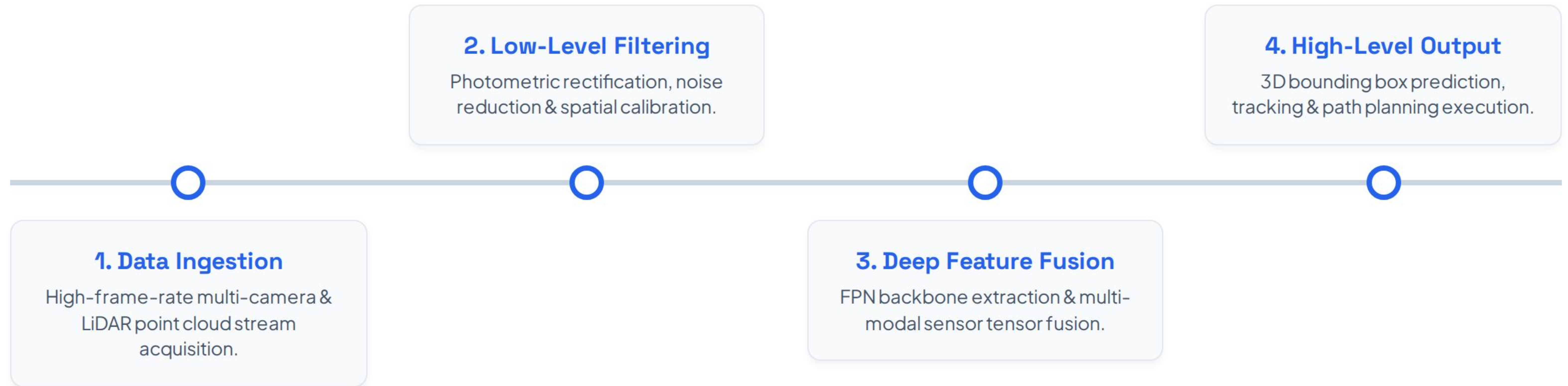
Engineering Context & Constraints

Self-driving vehicles navigate highly dynamic urban environments requiring real-time obstacle detection under strict latency constraints (Inference $\leq 33\text{ms}$ at 30 FPS).

Sensors must reliably detect pedestrians, road boundaries, and active traffic signals under adverse conditions including heavy glare, rain, and partial occlusions with zero tolerance for false negatives.



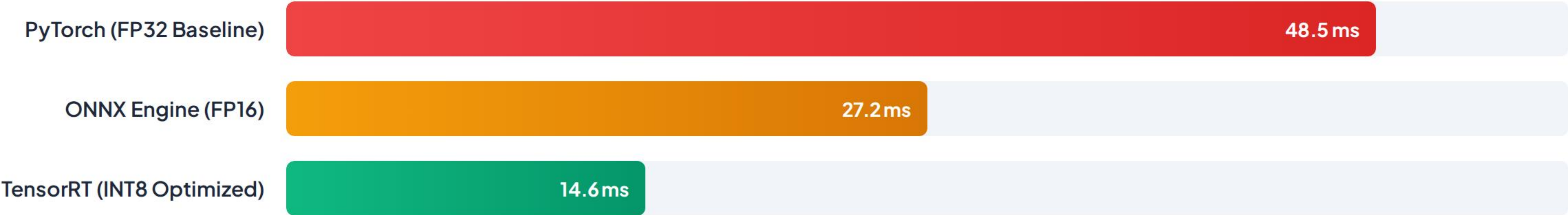
Perception Pipeline Flow



Performance Benchmarks

Framework Engine	Quantization Format	Inference Latency	mAP @ 0.5 (Objects)	Drivable Area IoU
PyTorch Baseline	FP32 (Standard)	48.5 ms	88.2%	93.1%
ONNX Runtime	FP16 (Half Precision)	27.2 ms	88.0%	93.0%
NVIDIA TensorRT	INT8 (Quantized)	14.6 ms	87.8%	92.8%
OpenCV DNN Module	FP32 (CPU Fallback)	112.0 ms	85.4%	89.5%

Model Optimization Latency



INT8 TensorRT quantization reduces execution latency below the critical 33ms frame window, enabling real-time 60 FPS safety control.

Ethical & Safety Governance

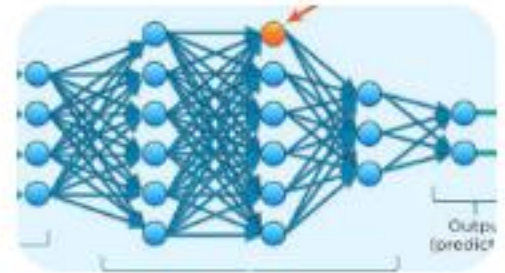
-  **Fail-Safe Thresholding:** Automated handover to secondary redundant sensor arrays if perception confidence drops below $\tau < 0.65$.
-  **Dataset Bias Mitigation:** Enforcing balanced demographic representation across lighting and weather variations to prevent systematic failure modes.
-  **Privacy Preservation:** Edge-based blurring of facial features and license plate tags prior to telemetry transmission.
-  **ISO 26262 Compliance:** Meeting automotive functional safety standards (ASIL-D) for continuous software verification.

Questions & Discussion

Thank you for your attention

Computer Vision & Autonomous Perception Research Group

Image Sources



https://daxg39y63pxwu.cloudfront.net/images/blog/deep-learning-architectures/Deep_Learning_Architecture_Diagram__by_ProjectPro.webp

Source: www.projectpro.io



<https://annotationbox.com/wp-content/uploads/2025/11/Bounding-Box-for-Object-Detection.webp>

Source: annotationbox.com